

PART-A

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INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	6	9	20

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$Y_1 = \theta_1 + \theta_2 + \epsilon_1,$$

$$Y_2 = \theta_1 - 2\theta_2 + \epsilon_2,$$

$$Y_3 = 2\theta_1 - \theta_2 + \epsilon_3,$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

~~X~~ from the eqn's

$$X = \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix}$$

and we know, $\hat{\theta} = (X^T X)^{-1} X^T y$

$$X^T X = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}$$

$$(X^T X)^{-1} = \frac{1}{27} \begin{bmatrix} 6 & 3 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 2/9 & 1/9 \\ 1/9 & 2/9 \end{bmatrix}$$

$$\begin{aligned}
 (X^T X)^{-1} X^T &= \begin{bmatrix} 2/9 & 1/9 \\ 1/9 & 2/9 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix}_{2 \times 3} \\
 &= \begin{bmatrix} 1/3 & 0 & 1/3 \\ 1/3 & -1/3 & 0 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 \hat{\theta} &= (X^T X)^{-1} X^T y = \begin{bmatrix} 1/3 & 0 & 1/3 \\ 1/3 & -1/3 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \\
 &= \begin{bmatrix} 1/3 y_1 + 1/3 y_3 \\ 1/3 y_1 - 1/3 y_2 \end{bmatrix}
 \end{aligned}$$

$$\Rightarrow \hat{\theta}_1 = \frac{y_1}{3} + \frac{y_3}{3}, \hat{\theta}_2 = \frac{y_1}{3} - \frac{y_2}{3}$$

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$$\cancel{E(\hat{\theta}_1) = E\left(\frac{y_1 + y_3}{3}\right) = 0, E(\hat{\theta}_2) = 0}$$

$$\cancel{Var(\hat{\theta}_1) =}$$

2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.
- (a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.
- (b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

$$(a) \quad X_1 \sim U_1, X_2 \sim U_2, X_3 \sim U_1 + U_2$$

$$U_1, U_2 \rightarrow \text{independent} \Rightarrow \text{Cov}(U_1, U_2) = 0$$

$$\text{Cov}(X_1, X_2) = \text{Cov}(U_1, U_2) = 0$$

$$\begin{aligned} \text{Cov}(X_1, X_3) &= \text{Cov}(U_1, U_1 + U_2) = \text{Cov}(U_1, U_1) + \text{Cov}(U_1, U_2) \\ &= \text{Var}(U_1) \end{aligned}$$

$$\Rightarrow \text{Cov}(X_2, X_3) = \text{Var}(U_2)$$

~~$$\text{Var}(U_1) = \frac{1}{12}$$~~

pdf for U_1, U_2

$$\begin{aligned} \text{Var}(X_3) &= \text{Var}(U_1) + \text{Var}(U_2) \\ &= \frac{1}{6} \end{aligned}$$

$$X \sim U(0,1) \quad f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

$$E[X] = \int_0^1 x dx = \frac{1}{2}$$

$$E[X^2] = \int_0^1 x^2 dx = \frac{1}{3} \Rightarrow \text{Var}(X) = \frac{1}{3} - \frac{1}{4} = \boxed{\frac{1}{12}}$$

$$\rho(X_1, X_2) = \frac{\text{Cov}(X_1, X_2)}{\sqrt{\text{Var}(X_1)} \sqrt{\text{Var}(X_2)}} = 0$$

$$\rho(X_1, X_3) = \frac{1/12}{\sqrt{1/12} \sqrt{1/6}} = 1/\sqrt{2}$$

$$\rho(X_2, X_3) = \frac{1/12}{\sqrt{1/12} \sqrt{1/6}} = 1/\sqrt{2}$$

03 $\rho = \begin{bmatrix} 1 & 0 & 1/\sqrt{2} \\ 0 & 1 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1 \end{bmatrix} \rightarrow \text{correlation matrix for } X$

(b) $\rightarrow$ last page

(b) Eigen Values

$$\Rightarrow \begin{vmatrix} 1-\lambda & 0 & 1/\sqrt{2} \\ 0 & 1-\lambda & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)((1-\lambda)^2 - 1) + 1(\lambda - 1) = 0$$

$$(1-\lambda)[(1-\lambda)^2 - 2] = 0$$

$$\Rightarrow \lambda = 1, 1-\sqrt{2}, 1+\sqrt{2}$$

$\lambda_2 \quad \lambda_3 \quad \lambda_1$

Eigen vectors

for λ_1

$$\begin{bmatrix} -\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & -\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & -\sqrt{2} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = 0$$

$$\Rightarrow -\sqrt{2}a_1 + \frac{a_3}{\sqrt{2}} = 0$$

$$-\sqrt{2}a_2 + \frac{a_3}{\sqrt{2}} = 0$$

$$\frac{a_1 + a_2}{\sqrt{2}} - \sqrt{2}a_3 = 0$$

$$\Rightarrow a_1 : a_2 : a_3 = 1 : 1 : \sqrt{2}$$

$$\Rightarrow e_1 = \begin{bmatrix} 1/2 \\ 1/2 \\ 1/\sqrt{2} \end{bmatrix} \text{ (after normalizing)}$$

Variance of 1st principal comp. $= 1+\sqrt{2}$

$$a_1 = a_3/\sqrt{2}$$

$$a_2 = a_3/\sqrt{2}$$

$$a_3 = \frac{a_1 + a_2}{\sqrt{2}} = \sqrt{2}a_1$$

$$a_3 =$$

$$1 : 1 : \sqrt{2}$$

$$1/\sqrt{6} \quad 1+1+2$$

$$1/\sqrt{6}$$

$$\sqrt{2}/\sqrt{6}$$

$$\frac{x_1}{\sqrt{6}} + \frac{x_2}{\sqrt{6}} + \frac{x_3}{\sqrt{6}}$$

$$\frac{1}{\sqrt{6}} + \frac{1}{\sqrt{6}} + \frac{1}{\sqrt{6}}$$

$$+ \frac{1}{\sqrt{6}} + \frac{1}{\sqrt{6}}$$

(ctd at the end)

$$\frac{1}{6} \left(\frac{5}{6} \right) = \frac{5}{36}$$

3. Let the matrix of distance be

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

(a) (6 points) Cluster the four items using each of the following procedures:

- Complete linkage hierarchical procedure.
- Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

(a) (i) complete linkage

$$\Rightarrow D_{u,vw} = \max(D_{u,v}, D_{u,w})$$

while removing the row and column with least distance

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

merge (1,2)

	(12)	3	4
(12)	0		
3	11	0	
4	5	3.5	0

merge (3,4)

$$D_{3,(12)} = \max(11, 2) = 11$$

$$D_{4,(12)} = \max(5, 3) = 5$$

$$D_{(34),(12)} = \max(11, 5)$$

	(12)	(34)
(12)	0	
(34)	11	0

↓
final

(ii) Average Linkage

$$D_{u,(vw)} = \frac{D_{u,v} + D_{u,w}}{2}$$

	1	2	3	4
1	0			
2	①	0		
3	11	2	0	
4	5	3	3.5	0

merge(1,2) →

	(12)	3	4
(12)	0		
3	6.5	0	
4	4	③-5	0

$$D_{3,(12)} = \frac{11 + 2}{2} = 6.5$$

$$D_{4,(12)} = \frac{5 + 3}{2} = 4$$

merge(3,4)

$$D_{(12),(34)} = \frac{D_{(12),3} + D_{(12),4}}{2}$$

$$= \frac{6.5 + 4}{2} = 5.25$$

	(12)	(34)
(12)	0	
(34)	5.25	0

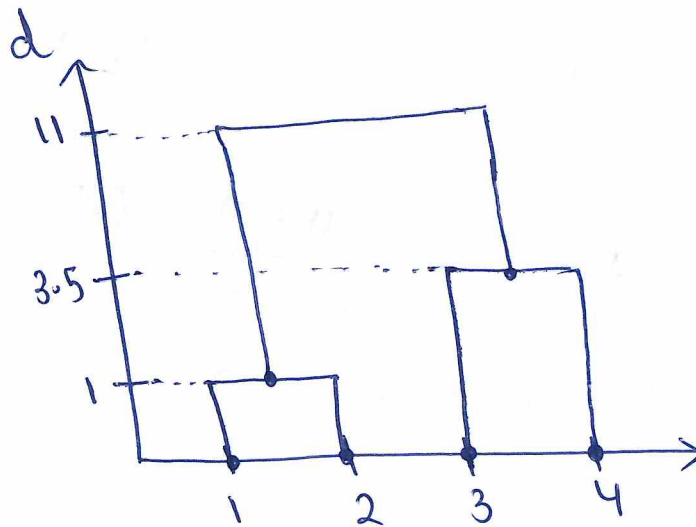
↓
final matrix

06

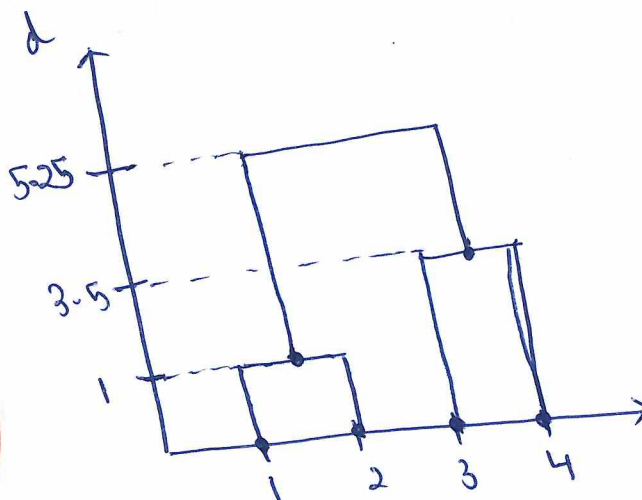
Q.3(b)

Dendrograms

(i) Complete Linkage



(ii) Average Linkage



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in both procedures, we get same sequence of clusters
but (12) and (34) in the end are at different
distances (1 vs 5.25)

Q.2 (b) (ctd)

for λ_2

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = 0$$

$$a_3 = 0$$

$$a_1 + a_2 = 0 \Rightarrow e_2 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix}$$

for λ_3

$$\begin{bmatrix} \sqrt{2} & 0 & 1 \\ 0 & \sqrt{2} & 1 \\ 1 & 1 & \sqrt{2} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = 0$$

$$\Rightarrow \sqrt{2}a_1 + a_3 = 0$$

$$\sqrt{2}a_2 + a_3 = 0$$

$$a_1 + a_2 + \sqrt{2}a_3 = 0$$

$$a_1 : a_2 : a_3 = 1 : 1 : -\sqrt{2}$$

$$\Rightarrow e_3 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

first comp.

$$y_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} [x_1 \ x_2 \ x_3] = \frac{x_1}{2} + \frac{x_2}{2} + \frac{x_3}{\sqrt{2}}$$

$$y_2 = \frac{x_1}{\sqrt{2}} - \frac{x_2}{\sqrt{2}}, \quad y_3 = \frac{x_1}{2} + \frac{x_2}{2} - \frac{x_3}{\sqrt{2}}$$

Q.2(b)

eigen values

$$\begin{vmatrix} 1-\lambda & 0 & 1/\sqrt{2} \\ 0 & 1-\lambda & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)((1-\lambda)^2 - \frac{1}{2}) + \frac{1}{\sqrt{2}} \left(\frac{-(1-\lambda)}{\sqrt{2}} \right)$$

$$= (1-\lambda)((1-\lambda)^2 - 1) = 0$$

$$\Rightarrow \boxed{\lambda = 1, 0, 2}$$

 $\lambda_1 = 2$ (largest)

↓

variance of first principal component

 (66.67%)

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